Hansi Zeng

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Education

University of Massachusetts Amherst Sep 2021 – Present
Ph.D. in Computer Science
- **Advisor**: Hamed Zamani; **Thesis**: Generative Information Retrieval for the Real World

University of Utah Sep 2019 – May 2021
M.S. in Computer Science
- **Advisor**: Qingyao Ai

University of Wisconsin–Madison Sep 2017 – Jun 2019
M.S. in Mathematics

Nankai University Sep 2014 – Jun 2018
B.S. in Mathematics

Research Interests

Information Retrieval, Large Language Models, Reinforcement Learning

Experience

Snapchat Sep 2025 – Present
Part-time Research Intern
- Identified retrieval as a key bottleneck in agentic search by investigating the role of retrieval systems in search agents
- Developed a dual-agent reinforcement learning framework to jointly optimize reasoning and document ranking for agentic search

Amazon May 2025 – Aug 2025
Applied Scientist Intern
- Studied web research agents, focusing on how tool exploration shapes reinforcement learning for deep research
- Developed synthetic planning methods that improve reinforcement learning for tool use and enhance multi-hop and open-web reasoning

Google DeepMind May 2024 – Mar 2025
Student Research Intern
- Studied prediction of LLM downstream performance for model selection across data mixtures, training objectives, and checkpoints
- Developed lightweight proxies and a learning-to-compare framework for ranking model variants without full fine-tuning

Amazon May 2023 – Dec 2023
Applied Scientist Intern
- Developed a novel generative information retrieval algorithm that scales to large-scale IR datasets, resulting in a first-author paper at WWW 2024
- Developed an efficient decoding algorithm for generative retrieval, resulting in a first-author paper at SIGIR 2024

Lowe's May 2022 – Aug 2022
Data Science Graduate Intern
- Studied personalized dense retrieval for unified information access across recommender systems and information retrieval, resulting in a first-author paper at SIGIR 2023

Selected Publications *(Citations: 1441, h-index: 13, i10-index: 14, as of Apr 2026)*

CoSearch: Joint Training of Reasoning and Document Ranking via Reinforcement Learning for Agentic Search
Hansi Zeng, Liam Collins, Bhuvesh Kumar, Neil Shah, Hamed Zamani
(In Submission) [Paper] [Code]

LLM Development at Fixed Scale: Lightweight Proxies to Overcome the Perplexity Illusion
Hansi Zeng, Kai Hui, Honglei Zhuang, Zhen Qin, Zhenrui Yue, Hamed Zamani, Dana Alon
(In Submission)

SynPlanResearch-R1: Encouraging Tool Exploration for Deep Research with Synthetic Plans
Hansi Zeng, Zoey Li, Yifan Gao, Chenwei Zhang, Xiaoman Pan, Tao Yang, Fengran Mo, Jiacheng Lin, Xian Li, Jingbo Shang
(In Submission) [Paper] [Code]

Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning
Bowen Jin, **Hansi Zeng**, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, Jiawei Han
COLM 2025 [Paper] [Code]

Scaling Sparse and Dense Retrieval in Decoder-Only LLMs
Hansi Zeng, Julian Killingback, Hamed Zamani
SIGIR 2025 [Paper] [Code]

Hypencoder: Hypernetworks for Information Retrieval
Julian Killingback, **Hansi Zeng**, Hamed Zamani
SIGIR 2025 [Paper] [Code]

Inference Scaling for Long-Context Retrieval Augmented Generation
Zhenrui Yue, Honglei Zhuang, Aijun Bai, Kai Hui, Rolf Jagerman, **Hansi Zeng**, Zhen Qin, Dong Wang, Xuanhui Wang, Michael Bendersky
ICLR 2025 (Oral) [Paper]

Planning Ahead in Generative Retrieval: Guiding Autoregressive Generation through Simultaneous Decoding
Hansi Zeng, Chen Luo, Hamed Zamani
SIGIR 2024 [Paper] [Code]

Scalable and Effective Generative Information Retrieval
Hansi Zeng, Chen Luo, Bowen Jin, Sheikh Muhammad Sarwar, Tianxin Wei, Hamed Zamani
WWW 2024 (Oral) [Paper] [Code]

Language Models as Semantic Indexers
Bowen Jin, **Hansi Zeng**, Guoyin Wang, Xiusi Chen, Tianxin Wei, Ruirui Li et al.
ICML 2024 [Paper] [Code]

Soft Prompt Decoding for Multilingual Dense Retrieval
Zhiqi Huang, **Hansi Zeng**, Hamed Zamani, James Allen
SIGIR 2023 [Paper]

A Personalized Dense Retrieval Framework for Unified Information Access
Hansi Zeng, Surya Kallumadi, Zaid Alibadi, Rodrigo Nogueira, Hamed Zamani
SIGIR 2023 [Paper]

Curriculum Learning for Dense Retrieval Distillation **(Best Short Paper Award)** 🏆
Hansi Zeng, Hamed Zamani, Vishwa Vinay
SIGIR 2022 [Paper] [Code]

A Zero Attentive Relevance Matching Network for Review Modeling in Recommender System
Hansi Zeng, Zhichao Xu, Qingyao Ai
ECIR 2021 [Paper]

Teaching

- COMPSCI 646: Information Retrieval, UMass Amherst — Spring 2025

- ECE 3530: Engineering Probability and Statistics, University of Utah — Fall 2020

Awards

- Best Short Paper Award, SIGIR — 2022
- Second-class Scholarship, Nankai University — 2015–2016

Service

Reviewer at SIGIR, WWW, COLM, ICML, NeurIPS, TOIS, CIKM, ACL, ECIR